Thermal Feedback in a Microscopic Black Hole Reactor

Stability of a coupled Hawking–Bondi model

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Abstract

In a fixed medium, the balance between idealized Bondi accretion and blackbody Hawking evaporation of a microscopic black hole is unstable. We ask whether heating of the medium by the hole's own radiation can stabilize it. The black-hole mass is coupled to an effective temperature of the surrounding gas: the ideal-gas sound speed makes the Bondi coefficient scale as $T^{-3/2}$, while deposited Hawking power and linear cooling set the temperature. The resulting two-variable system has exactly one positive equilibrium, hotter and more massive than the fixed-medium balance. Its Jacobian determinant is negative for every positive heating and cooling strength, so the equilibrium is a saddle. The same holds for any one-zone model in which heating decreases with mass, accretion decreases with temperature and cooling increases with temperature. Thermal coupling raises the growth rate of perturbations relative to a frozen temperature at the same equilibrium, although the shifted equilibrium departs more slowly than the fixed-medium one. Order-of-magnitude estimates for two hydrogen plasmas show that neither meets the conditions under which a one-zone thermal reservoir would describe the flow at the balance mass.

1 Introduction

In a previous paper [1] we balanced idealized Bondi accretion against a blackbody approximation to Hawking evaporation,

$$\dot{M} = AM^2 - \frac{B}{M^2}, \qquad A, B > 0, \tag{1}$$

and showed that its only positive balance, $M_{c0} = (B/A)^{1/4}$, is unstable: the right-hand side increases with mass, so a perturbation of either sign grows. In dimensionless form every case reduces to $dx/ds = x^2 - x^{-2}$, whose fixed point $x = 1$ has growth rate 4.

That analysis held the density and sound speed of the medium fixed. This is a strong restriction, because the hole is itself a powerful heat source: at the reference balance mass of Ref. [1], its blackbody luminosity is 3.6×10^{11} W. Heating raises the sound speed, and the Bondi rate falls as its inverse cube. The question of the present paper is therefore:

> *Can heating of the accreting medium by Hawking radiation turn the repelling balance into an attracting one?*

Recent time-dependent calculations of primordial black holes accreting inside stars show that the thermodynamics of the inflowing gas can change the classical adiabatic Bondi rate substantially [6]. Their luminosity comes from the accreting gas rather than from Hawking emission, so we take from them only the modeling principle: the thermal state of the medium should be allowed to evolve and feed back on accretion.

We add the smallest ingredient that makes this question testable: one effective temperature $T(t)$ of the medium, evolving together with $M(t)$ at fixed density. Section 2 sets up the model and Sec. 3 locates its equilibrium. Section 4 proves that the equilibrium is a saddle for all positive parameters, compares three growth rates that are easily confused, and extends the result to a class of monotone heating and cooling laws. Section 5 checks these statements numerically, and Sec. 6 estimates whether any simple plasma could support the one-zone description.

Throughout, the black hole is uncharged and nonrotating, the emission follows the photon-blackbody baseline of Ref. [1], and "stability" refers to the equilibrium of the stated differential equations. The word "reactor" names the motivating question of this series; confinement, energy collection and engineering feasibility are outside its scope.

2 Model

2.1 Accretion and evaporation

For a Schwarzschild black hole the Hawking temperature scales as $T_H \propto M^{-1}$ [2]. Treating the horizon as a photon blackbody gives

$$P_{\rm bb} = \frac{K}{M^2}, \qquad K = \frac{\hbar c^6}{15360\pi G^2}, \qquad \dot{M}_{\rm evap} = -\frac{B}{M^2}, \qquad B = \frac{K}{c^2}.$$

This is a simplified baseline: greybody factors and the additional particle species available at high temperature change the emission rate [3, 4].

Steady spherical Bondi accretion of a gas at rest far from the hole gives [5]

$$\dot{M}_{\rm acc} = A\,M^2, \qquad A = \frac{4\pi\lambda\rho_\infty G^2}{c_s^3}.$$

For an ideal gas with fixed composition and adiabatic index, $c_s^2 = \gamma k_B T/(\mu m_p)$. At fixed density ρ_∞ and accretion parameter λ, the coefficient therefore depends on temperature as

$$A(T) = A_0 \left(\frac{T}{T_0}\right)^{-3/2}. \tag{2}$$

The temperature T must describe the gas that sets the Bondi boundary condition, at radii of order the Bondi radius and beyond. Heating only a thin layer near the horizon would not change that boundary condition. We therefore assume a well-mixed reservoir in which deposited radiation reaches the gas that controls accretion. This is a closure assumption; Sec. 6 examines when it could hold.

The water-like reference medium of Ref. [1] is not an ideal gas and does not define a consistent T_0. We use its balance mass and time unit only as scales and calibrate no thermal parameter to water.

2.2 Heating and cooling

Let a fraction ϵ of the blackbody luminosity be deposited in a reservoir of constant heat capacity $C_{\rm env}$, and let a linear cooling law return the reservoir to T_0 on a timescale τ_c. The model is

$$\dot{M} = A_0 \left(\frac{T}{T_0}\right)^{-3/2} M^2 - \frac{B}{M^2}, \qquad C_{\rm env}\dot{T} = \frac{\epsilon K}{M^2} - \frac{C_{\rm env}}{\tau_c}(T - T_0), \tag{3}$$

with all constants positive, $0 < \epsilon \leq 1$ and $M, T > 0$. The parameters ϵ, $C_{\rm env}$ and τ_c are not derived from the emission spectrum, opacity or geometry. The reservoir has no spatial structure and does not track the energy carried by gas flowing in and out.

2.3 The feedback loop

Equation (3) contains a single closed loop (Fig. 1). A decrease of M raises $P_{\rm bb}$; the deposited power heats the reservoir; the higher sound speed lowers $A(T)$; and the weaker accretion lets evaporation remove mass faster. The loop therefore reinforces the initial perturbation. A complete description would also include the response of the density and a transport delay between the deposition region and the Bondi radius. Neither is modeled here: holding ρ_∞ fixed isolates the sound-speed channel.

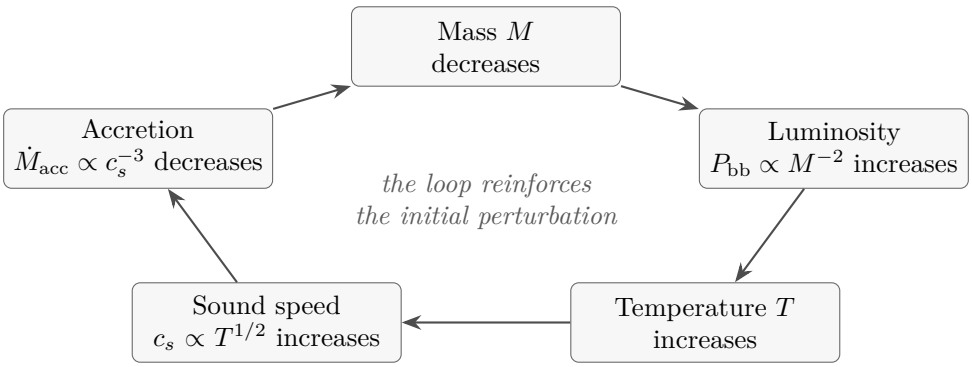


Figure 1: The thermal feedback loop for a downward mass perturbation. Every link follows from Eq. (3). An upward perturbation runs the same loop with every change reversed and favors further growth.

2.4 Dimensionless form

Let $M_{c0} = (B/A_0)^{1/4}$ be the fixed-medium balance mass at $T = T_0$ and $t_* = (A_0 M_{c0})^{-1}$ its time unit. With $x = M/M_{c0}$, $\theta = T/T_0$ and $s = t/t_*$, Eq. (3) becomes

$$\frac{dx}{ds} = F(x,\theta) = \theta^{-3/2}x^2 - x^{-2}, \qquad \frac{d\theta}{ds} = G(x,\theta) = \eta x^{-2} - \zeta(\theta - 1), \tag{4}$$

where

$$\eta = \frac{t_* \epsilon K}{C_{\rm env} T_0 M_{c0}^2}, \qquad \zeta = \frac{t_*}{\tau_c}, \qquad \chi = \frac{\eta}{\zeta} = \frac{\epsilon P_{\rm bb}(M_{c0})\,\tau_c}{C_{\rm env} T_0}. \tag{5}$$

The feedback strength χ compares the energy deposited during one cooling time with the thermal energy scale $C_{\rm env}T_0$ of the reservoir. The cooling rate ζ compares the cooling time with the mass time unit of the fixed-medium model. The two can be varied independently; changing τ_c alone changes both.

3 The shifted equilibrium

The nullclines of Eq. (4) are

$$F = 0: \; x = \theta^{3/8}, \qquad G = 0: \; \theta = 1 + \chi x^{-2}.$$

The first rises with temperature and the second falls with mass, so they intersect once. The intersection (x_*, θ_*) satisfies

$$x_* = \theta_*^{3/8}, \qquad \theta_* - 1 = \chi\, \theta_*^{-3/4}. \tag{6}$$

The function $h(\theta) = \theta - 1 - \chi\theta^{-3/4}$ is negative at $\theta = 1$, grows without bound, and has $h'(\theta) = 1 + \frac{3}{4}\chi\theta^{-7/4} > 0$. Hence for every $\chi > 0$ there is exactly one positive equilibrium, with $\theta_* > 1$ and $x_* > 1$. Its position depends on χ alone. In the two limits,

$$\chi \ll 1: \; \theta_* = 1 + \chi + O(\chi^2), \quad x_* = 1 + \tfrac{3}{8}\chi + O(\chi^2); \qquad \chi \gg 1: \; \theta_* \sim \chi^{4/7}, \quad x_* \sim \chi^{3/14}.$$

Heating therefore moves the balance to a hotter medium and a larger mass (Fig. 2a). Whether trajectories return to the shifted balance is a separate question, answered next.

4 Linear stability

4.1 A saddle for every positive coupling

The partial derivatives of Eq. (4) are

$$F_x = 2\theta^{-3/2}x + 2x^{-3}, \quad F_\theta = -\tfrac{3}{2}\theta^{-5/2}x^2, \quad G_x = -2\eta x^{-3}, \quad G_\theta = -\zeta.$$

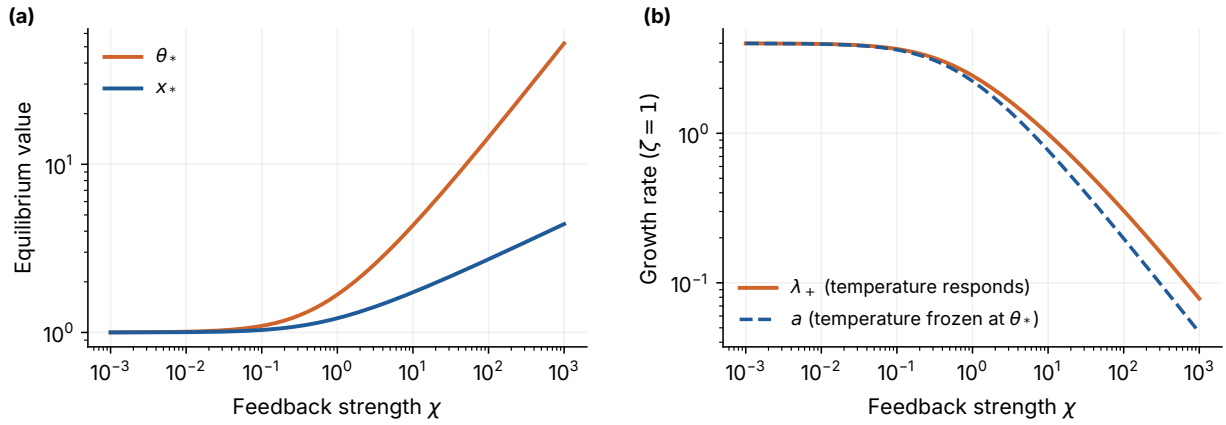


Figure 2: Heating shifts the balance without making it attracting. (a) Equilibrium temperature and mass from Eq. (6). (b) For $\zeta = 1$, the unstable eigenvalue λ_+ of the coupled system and the growth rate $a = 4x_*^{-3}$ obtained when the temperature is frozen at θ_*. Both rates fall as the equilibrium moves to larger mass, but $\lambda_+ > a$ throughout.

At the equilibrium $\theta_*^{-3/2} = x_*^{-4}$, so the Jacobian is

$$J_* = \begin{pmatrix} a & b \\ c & -\zeta \end{pmatrix}, \qquad a = 4x_*^{-3} > 0, \quad b = -\tfrac{3}{2}\theta_*^{-5/2}x_*^2 < 0, \quad c = -2\eta x_*^{-3} < 0,$$

with determinant

$$\det J_* = -4\zeta x_*^{-3} - 3\eta\,\theta_*^{-5/2}x_*^{-1} < 0. \tag{7}$$

A real 2×2 matrix with negative determinant has two real eigenvalues of opposite sign. The equilibrium is therefore a saddle for every $\eta, \zeta > 0$. A stable node, a stable focus and a Hopf bifurcation are all excluded, because each requires $\det J_* > 0$.

4.2 Three growth rates

Three rates are easily confused when judging the effect of heating. The first is a, the growth rate of a mass perturbation at the new equilibrium when the temperature is frozen at θ_*. The second is the unstable eigenvalue of the coupled system,

$$\lambda_\pm = \frac{a - \zeta \pm \sqrt{(a+\zeta)^2 + 4bc}}{2}, \qquad \lambda_+ > a > 0, \qquad \lambda_- < -\zeta, \tag{8}$$

where the inequalities follow from $bc > 0$. The third is the rate 4 of the fixed-medium balance. For $\chi = \zeta = 1$ these are $a = 2.234$, $\lambda_+ = 2.431$ and 4. Letting the temperature respond raises the growth rate at a given equilibrium from a to λ_+: the coupling itself is destabilizing. The shift of the equilibrium to larger mass lowers all rates, which is why λ_+ is smaller than 4. A smaller λ_+ therefore reflects the shifted equilibrium, not a stabilizing effect of the feedback.

The unstable eigenvector has slope $(\lambda_+ - a)/b < 0$ in the (x, θ) plane and the stable one has slope $(\lambda_- - a)/b > 0$. Only initial conditions on the one-dimensional stable manifold approach the equilibrium; a generic perturbation has an unstable component. A loss of mass at the equilibrium temperature, for example, raises the heating term and lowers the accretion coefficient, both of which accelerate further loss.

4.3 Limiting cases

For fixed ζ and $\chi \gg 1$, Eq. (8) gives $\lambda_+ \sim 7\chi^{-9/14}$ (Appendix A). The growth rate becomes small but stays positive for every finite χ. At fixed χ, changing ζ also changes $\eta = \chi\zeta$; the slow- and fast-cooling limits are

$$\lim_{\zeta \to 0^+} \lambda_+ = a, \qquad \lim_{\zeta \to \infty} \lambda_+ = a + 3\chi\,\theta_*^{-5/2}x_*^{-1} > a.$$

Table 1: Equilibria, growth rates and Jacobian determinants for $\zeta = 1$. The rate a freezes the temperature at θ_*; $\lambda_\pm$ include the temperature response. All quantities are dimensionless.

χ	θ_*	x_*	a	λ_-	λ_+	$\det J_*$
0.01	1.00993	1.00371	3.95580	−1.00588	3.96168	−3.98496
0.1	1.09352	1.03409	3.61728	−1.04971	3.66699	−3.84929
1	1.67821	1.21428	2.23413	−1.19734	2.43146	−2.91128
10	4.33092	1.73268	0.76897	−1.22271	0.99167	−1.21253
100	14.47513	2.72417	0.19786	−1.10595	0.30381	−0.33600

In the first limit the temperature cannot respond during a mass e-folding; in the second it follows the mass adiabatically along $G = 0$. Neither limit reverses the instability.

4.4 A sign theorem

The saddle does not depend on the specific power laws or on linear cooling.

Proposition 1. *Consider the one-zone system*

$$\dot{M} = \mathcal{F}(M,T) = A(T)M^2 - BM^{-2}, \qquad \dot{T} = \mathcal{G}(M,T) = H(M) - \Lambda(T),$$

with $B > 0$ and differentiable A, H, Λ. If at a positive equilibrium $A > 0$, $A'(T) < 0$, $H'(M) < 0$ and $\Lambda'(T) > 0$, that equilibrium is a saddle.

Proof. At the equilibrium, $\mathcal{F}_M = 2AM + 2BM^{-3} > 0$, $\mathcal{F}_T = A'M^2 < 0$, $\mathcal{G}_M = H' < 0$ and $\mathcal{G}_T = -\Lambda' < 0$. Hence $\det D(\mathcal{F},\mathcal{G}) = \mathcal{F}_M\mathcal{G}_T - \mathcal{F}_T\mathcal{G}_M < 0$. □

The three conditions have direct physical meanings: hotter gas accretes less efficiently, a lighter hole heats its surroundings more strongly, and a hotter reservoir cools faster. They also cover a medium held at fixed pressure rather than fixed density. Then $\rho_\infty \propto T^{-1}$, so $A \propto T^{-5/2}$ falls even faster with temperature and the proposition still applies. Stabilization within a one-zone description would require reversing one of these signs. A model with additional state variables, such as an independently evolving density, is not covered by the proposition, and its stability has to be determined from its own Jacobian.

5 Numerical checks

5.1 Method

We solve Eq. (6) with a bracketed root finder, evaluate J_* analytically, and integrate Eq. (4) with `scipy.integrate.solve_ivp` (DOP853, relative tolerance 10^{-10}, absolute tolerance 10^{-12}, maximum step 0.01). Trajectories are stopped at the edges of the plotted domain, well away from the singular boundary $x = 0$. For $\chi = \zeta = 1$,

$$(x_*, \theta_*) = (1.214276,\ 1.678211), \qquad (\lambda_-, \lambda_+) = (-1.197336,\ 2.431462).$$

Table 1 lists the equilibrium and the rates for five feedback strengths. In every row $\lambda_+ > a > 0$, while both rates decrease as the equilibrium moves to larger mass.

5.2 Phase portrait

Figure 3 shows the nullclines and the manifolds of the saddle for $\chi = \zeta = 1$. The manifolds are integrated from displacements of 10^{-6} along the eigenvectors, backward in time for the stable one and forward for the unstable one. Trajectories to the right of the stable manifold gain mass and those to its left lose it; the figure shows the local geometry and does not determine global basins.

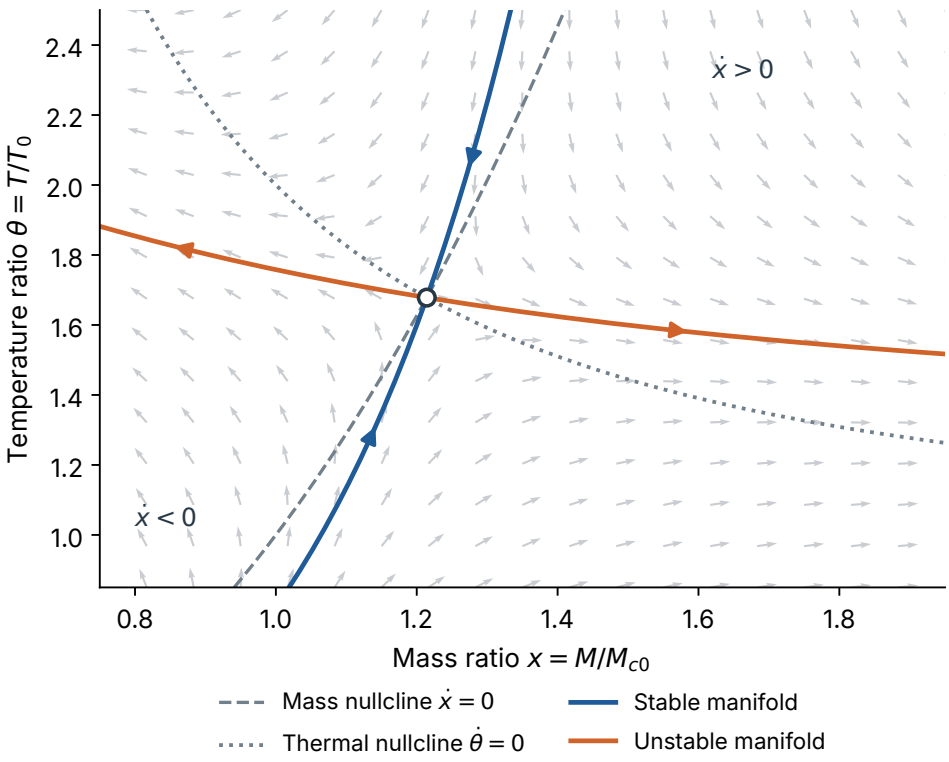


Figure 3: Phase portrait for $\chi = \zeta = 1$. The mass and thermal nullclines meet at the single equilibrium (open circle). Arrowheads mark the direction of time: the stable manifold (blue) approaches the saddle and the unstable manifold (orange) leaves it. Gray arrows show the direction of the flow.

5.3 Parameter scan

Figure 4 shows λ_+ on a 201×201 logarithmic grid with $10^{-2} \le \chi, \zeta \le 10^2$. It ranges from 0.20 to 3.98 and depends mainly on χ, which sets the equilibrium mass. On a separate 41×41 grid the script also confirms $\det J_* < 0$, $\lambda_+ > a > 0$ and $\lambda_- < -\zeta$ at every point. These scans illustrate Eq. (7); the analytic determinant, not the finite grid, establishes the result for all positive parameters.

5.4 Equal fractional perturbations

Figure 5 perturbs each model by $\pm 2\%$ in mass relative to its own equilibrium: the fixed-medium model from $x = 1 \pm 0.02$ and the coupled model from $(x_*(1 \pm 0.02), \theta_*)$, both on the time axis $s = t/t_*$. All four trajectories leave their balance. The coupled ones leave more slowly because the equilibrium mass is larger, in agreement with $\lambda_+ < 4$.

6 Physical scales

The saddle result does not depend on material constants, but identifying T with the temperature that sets the Bondi boundary condition does. We test this identification with two fully ionized hydrogen media: a hot plasma with $T_0 = 10^6$ K and proton density $n = 10^{20}\,\mathrm{m}^{-3}$, and a hydrogen medium at the central temperature and density of the Sun, $T_0 = 1.56 \times 10^7$ K and $\rho = 1.5 \times 10^5\,\mathrm{kg\,m}^{-3}$ [7]. The second case borrows only these two numbers and is not a model of the solar core. In both cases $\gamma = 5/3$, $\mu = 1/2$ and $\lambda = 1/4$.

6.1 Flow, heating and cooling times

At the balance mass M_{c0} we define the Bondi radius $r_B = GM_{c0}/c_s^2$ and the flow time $t_{\mathrm{flow}} = r_B/c_s$. A sphere of radius r_B contains thermal energy $E_B = 3nk_BT_0V_B$, with $V_B = 4\pi r_B^3/3$. The Thomson optical depth $\tau_T = n\sigma_T r_B$ estimates the probability that a photon scatters once while crossing it.

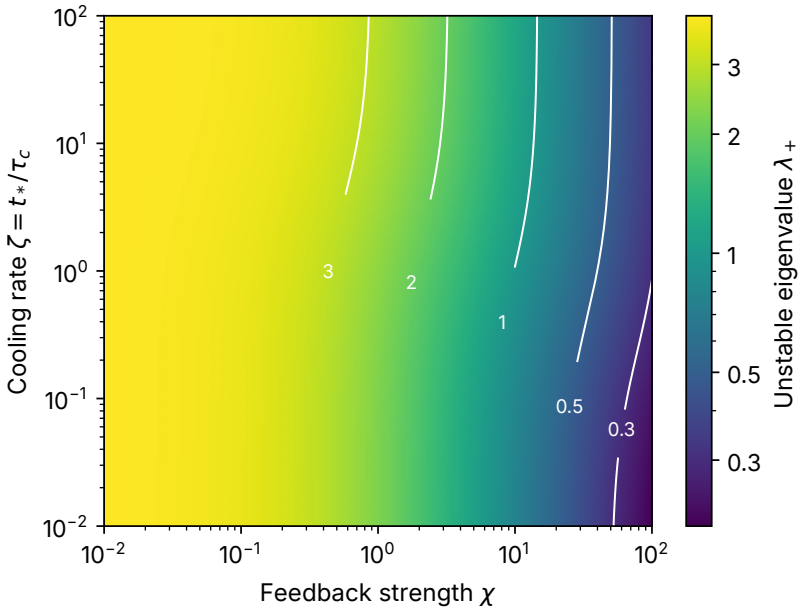


Figure 4: The unstable eigenvalue over the (χ, ζ) plane. White contours mark constant λ_+. The eigenvalue decreases with feedback strength but is positive everywhere, as Eq. (7) requires.

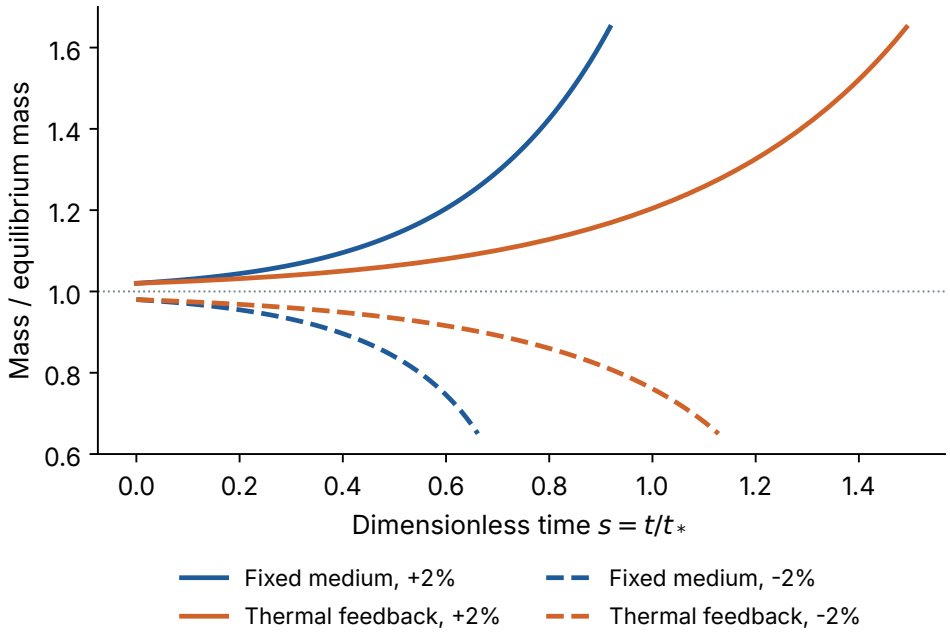


Figure 5: Both balances repel perturbations. Mass divided by each model's own equilibrium mass after $\pm 2\%$ perturbations; the coupled model has $\chi = \zeta = 1$ and starts at its equilibrium temperature. Curves end at plotting cutoffs.

Table 2: Order-of-magnitude scales at the fixed-medium balance mass M_{c0} for two fully ionized hydrogen media. Only T_0 and n are inputs; the other entries follow from the blackbody, Thomson, free–free and NRL collision prescriptions in the text.

Quantity	Hot plasma	Solar-center hydrogen
T_0 (K)	1.00×10^{6}	1.56×10^{7}
n (m^{-3})	1.00×10^{20}	8.97×10^{31}
c_s ($\mathrm{m\,s}^{-1}$)	1.66×10^{5}	6.55×10^{5}
M_{c0} (kg)	2.96×10^{14}	8.54×10^{11}
r_B (m)	7.19×10^{-7}	1.33×10^{-10}
t_{flow} (s)	4.34×10^{-12}	2.03×10^{-16}
τ_T	4.78×10^{-15}	7.92×10^{-7}
$\tau_T P_{\mathrm{bb}}(M_{c0})$ (W)	1.94×10^{-11}	3.87×10^{2}
$t_{\mathrm{heat},T}$ (s)	3.33×10^{-4}	1.47×10^{-15}
t_{ff} (s)	2.64	1.16×10^{-11}
ℓ_{ii}/r_B	2.1×10^{6}	3.1

If every scattered photon gave all of its energy to the gas, the heating rate would be $\tau_T P_{\mathrm{bb}}(M_{c0})$. This upper estimate gives the shortest possible photon heating time,

$$t_{\mathrm{heat},T} = \frac{E_B}{\tau_T P_{\mathrm{bb}}(M_{c0})}.$$

In reality a scattering transfers only part of the photon energy, and the cross section falls below the Thomson value at high energy. Other Hawking particles are not included in this estimate.

For cooling we use optically thin free–free emission of pure hydrogen from the NRL Plasma Formulary [8],

$$q_{\mathrm{ff}} = 1.69 \times 10^{-38}\, n^2 \sqrt{k_B T_0/(1\,\mathrm{eV})}\ \mathrm{W\,m}^{-3}, \qquad t_{\mathrm{ff}} = \frac{3 n k_B T_0}{q_{\mathrm{ff}}},$$

with n in m^{-3}. Free–free emission is only one cooling channel, and gas leaving a region of size r_B removes its energy on the time t_{flow} even if it radiates slowly. The ion mean free path $\ell_{ii} = v_{Ti}/\nu_{ii}$ uses the NRL thermal speed $v_{Ti} = (k_B T/m_p)^{1/2}$ and ion collision rate with $\ln\Lambda = 10$. Table 2 collects the results.

In the hot plasma, $t_{\mathrm{heat},T}/t_{\mathrm{flow}} \simeq 7.7 \times 10^{7}$; in the denser medium the ratio is still about 7. Free–free cooling is slower than the flow by factors of 6.1×10^{11} and 5.7×10^{4}. Gas therefore crosses the Bondi region long before it is heated or cools radiatively. If the exchange of gas relaxes the reservoir on $\tau_c \sim t_{\mathrm{flow}}$, Eq. (5) gives photon-only estimates $\chi \approx 1.3 \times 10^{-8}$ and 0.14, both upper limits. In addition, the ion mean free path exceeds r_B in both media, so even the continuum Bondi flow assumed at the balance mass is not established. The denser case is sensitive to composition, collision physics and the non-blackbody Hawking spectrum, so its numbers are diagnostics, not statements about the Sun.

6.2 Relation to accretion-powered feedback

Cantiello et al. [6] solve time-dependent spherical hydrodynamics with radiative cooling for primordial black holes in stellar cores. They identify a hot-Bondi regime, a bremsstrahlung-cooling regime that drives the flow toward isothermal, and a photon-trapping regime, and find that cooling raises the accretion rate by a factor of about 2–7 relative to adiabatic Bondi. Their black hole has no intrinsic luminosity; the radiation comes from the accreting gas.

This difference reverses the sign of the feedback. In an accretion-powered system, a higher accretion rate produces more radiation and more heating, which then limits accretion: the feedback is negative and tends to regulate the flow. A Hawking-powered system radiates more when its mass decreases, so the same thermal suppression of accretion acts in the direction of the original perturbation. This reversal is the physical origin of the saddle. For the same reason, their quantitative results cannot be used to fix ϵ, C_{env} or τ_c here.

Table 3: A hierarchy of Hawking–Bondi models.

Model	State variables	Accretion law
Fixed medium, Ref. [1]	M	$A = \text{const}$
Thermal feedback, this paper	M, T	$A(T)$
Density response	M, T, ρ	$A(T, \rho)$
Radiation hydrodynamics	M, $\rho(r)$, $T(r)$, $v(r)$	solved from the flow

6.3 Requirements for a physical interpretation

Connecting Eq. (4) to a real medium would require a collisional mean free path much smaller than r_B; heating and thermalization fast enough to change the sound speed at the Bondi radius; a deposition calculation for the actual Hawking spectrum; and an energy budget that includes advection, radiation and density evolution. A comparison of $t_{\rm heat}$, $t_{\rm cool}$ and $t_{\rm flow}$ is necessary but not sufficient. Accordingly, the illustrative choice $\chi = \zeta = 1$ in Sec. 5 is not a calibration of either medium in Table 2.

These requirements suggest a hierarchy of models in which each step adds one physical ingredient (Table 3).

The next step, a density response, adds a third state variable. Proposition 1 does not apply to it, so its stability has to be tested anew; a third variable may or may not change the outcome.

7 Conclusion

Coupling the Hawking–Bondi mass balance to a heated and cooled reservoir yields a single equilibrium, hotter and more massive than the fixed-medium balance. Its Jacobian determinant, Eq. (7), is negative for all positive feedback and cooling strengths, so the equilibrium is a saddle. The same holds for every one-zone model in which heating decreases with mass, accretion decreases with temperature and cooling increases with temperature. The thermal response is destabilizing: at a given equilibrium it raises the growth rate from a to λ_+. The lower absolute rate compared with the fixed-medium balance comes from the shift of the equilibrium to larger mass. The physical reason is the sign of the loop: a Hawking-powered hole radiates more as it loses mass, so heating suppresses accretion exactly when accretion is needed.

Timescale estimates for two hydrogen plasmas show that neither supports a one-zone thermal reservoir at its balance mass, because gas crosses the Bondi region faster than it heats or cools. Whether density evolution, spatial transport or realistic emission spectra can change the result remains open; the sign conditions of Proposition 1 identify what any such extension must check.

A Scaling and the strong-feedback limit

At M_{c0}, $B = A_0 M_{c0}^4$ and $t_*^{-1} = A_0 M_{c0}$. Substituting $M = M_{c0} x$, $T = T_0 \theta$ and $t = t_* s$ into Eq. (3) gives Eq. (4). The quantity $\epsilon K/(C_{\rm env} T_0 M_{c0}^2)$ has units of inverse time, so η and χ are dimensionless.

For $\chi \gg 1$, Eq. (6) gives $\theta_*^{7/4} \sim \chi$ and $x_* \sim \chi^{3/14}$, hence $a = 4x_*^{-3} \sim 4\chi^{-9/14}$. The off-diagonal product is $bc = 3\eta\, \theta_*^{-5/2} x_*^{-1} \sim 3\zeta\chi^{-9/14}$. At fixed ζ, $a \to 0$ and the positive root of Eq. (8) behaves as $\lambda_+ \sim a + bc/\zeta \sim 7\chi^{-9/14}$. This limit is not uniform if ζ also tends to zero.

B Reproducibility

The script `reproduce.py` requires NumPy, SciPy and Matplotlib. Running `python reproduce.py` in the paper directory computes the equilibria, Jacobians and eigenvalues, integrates the trajectories, evaluates Tables 1 and 2, regenerates Figs. 2–5 and writes `results.json`. Figure 1 is drawn in the LaTeX source; running `pdflatex main.tex` twice rebuilds the paper. Constants are SI

values: $G = 6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$, $\hbar = h/2\pi$ with $h = 6.62607015 \times 10^{-34}\,\mathrm{J\,s}$, $k_B = 1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$, $m_p = 1.67262192 \times 10^{-27}\,\mathrm{kg}$ and $\sigma_T = 6.6524587 \times 10^{-29}\,\mathrm{m^2}$.

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